

TOPS Programme 2026

Team 2

Adaptive Dual-Arm Robotics for Efficient CNC Automation

Smart coordination between a carrier robot and dual-arm manipulation

Team Members:

Shanmukha Vinayak M

Ayush Ranjan

Tanmay Patil

Motivation

- CNC machine tending requires **repetitive loading, unloading and handling**.
- **Conventional** industrial manipulators use a **single end-effector**, leading to sequential operations and **frequent repositioning**.
- Excessive carrier movement **increases cycle time**, energy consumption, and **reduces effective workspace** utilization.
- **Dual-arm manipulation** can perform more operations from the **same position**.

Objectives

- Design a **carrier-assisted** dual-arm manipulation system.
- **Reduce** unnecessary movements.
- **Improve** task efficiency and workspace usage
- **Validate** the framework in a **simulated CNC workcell**.



Proposed Framework

The proposed system consists of a carrier industrial manipulator equipped with a mounted dual-arm manipulation module for CNC machine tending.

- Carrier Manipulator
 - The carrier manipulator provides **global positioning** and moves the entire dual-arm module.
- Dual-Arm Module
 - Performs **local manipulation** within its reachable workspace.
- Smart Coordination
 - A hierarchical coordination strategy **determines** whether the **current task** can be completed locally or whether the carrier manipulator must reposition.

Goal: Minimize unnecessary carrier movement and improve CNC task efficiency.

AI generated image of proposed framework



Expected Deliverables

1. System Design

- Conceptual robotic architecture
- CAD model of the dual-arm manipulation module

2. Simulation Platform

- ROS2 & MoveIt integration
- URDF-based CNC workcell

3. Coordination Framework

- Hierarchical motion allocation
- Workspace-aware task execution

4. Performance Analysis

- Cycle time
- Carrier repositioning
- Workspace utilisation
- Task success rate

5. Documentation

- Technical report
- Project presentation
- Source code & simulation package

Project Timeline

Activity	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12
Literature Survey						
Architecture & CAD						
ROS2 & MoveIt						
Coordination Framework						
Simulation & Evaluation						
Documentation						

Team Responsibilities

Shanmukha Vinayak:

Build and integrate the complete **ROS2/MoveIt** robotic simulation environment, enabling **motion planning, robot modelling,** coordination, and simulation testing.

Ayush Ranjan:

Develop the **coordination layer** that optimizes task execution by **allocating motion** between the carrier robot and dual-arm module.

Tanmay Patil:

Design and validate the dual-arm module and **CNC workcell**, ensuring mechanical feasibility, **workspace compatibility**, and smooth integration with the overall simulation.

Expected Outcome

The proposed project is expected to deliver a **simulation-validated smart system** where the **main robot** and **two smaller arms** work together efficiently. It will **reduce** unnecessary robot movement, make **better use of the workspace**, and **improve** the overall task speed and performance.